16 Dec. 2020

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increment 63

ACME Advanced Combustion via Microgravity Experiments
s-Flame Structure and Response of Spherical Diffusion Flames

The s-Flame investigation is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The first round of s-Flame tests was conducted during Increments 60 and 61, while the second round was conducted fully within Increment 63. There were no tests conducted in Increment 62 because of limited crew availability.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for s-Flame and ACME information.

Principal Investigator (PI):	Prof. C.K. (Ed) Law, Princeton University, Princeton, NJ
Co-Investigator (Co-I):	Prof. Stephen D. Tse, Rutgers University, Piscataway, NJ
Lead collaborator:	Dr. Vladimir V. Gubernov, P.N. Lebedev Physical Institute of the Russian Academy of Sciences, Moscow, Russia
NASA Project Scientist (PS):	Dennis P. Stocker, NASA Glenn Research Center, Cleveland, OH

I. Introduction

Tests for the Structure and Response of Spherical Diffusion Flames (s-Flame) experiment were conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. The purpose of s-Flame is to advance the ability to predict the structure and dynamics, including extinction and instabilities, of both soot-free and sooty flames. The results may contribute to the development of lean-burn engines for improved efficiency and reduced pollutant emissions here on Earth.

The spherical flames are ignited at non-steady conditions and allowed to transition naturally towards steady-state or extinction. Tests are conducted with various inert diluents in both the fuel and chamber (oxidizer) atmosphere. The fuel gases include hydrogen and methane mixtures for soot-free flames, and ethylene for sooty flames. One experimental objective is to identify the extinction limits for both radiative and chemical kinetic extinction, i.e., where the flame extinguishes from radiative heat loss or is "blown out," respectively. Another objective is to determine the existence, onset, and nature of pulsating instabilities that have been theoretically predicted to occur in flames with certain fuel/inert mixtures at near-limit conditions.

More specifically, s-Flame has the following experimental objectives:

A. Measure and characterize the transient structure (such as temperature and species distribution) of diffusion flames established at initially non-steady, spherical flame structures for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty), for different flow rates.
B. Determine the quasi-steady convective/chemical-kinetic extinction limits (low system Damköhler numbers) and radiative/chemical-kinetic extinction limits (high system Damköhler numbers) for mixtures of different H_2/CH_4/diluent ratios (soot-free) and different C_2H_4/diluent ratios (sooty).
A. Determine the existence (for different Zeldovich number, Ze), onset (as a function Da), and nature (i.e. mode, frequency, amplitude) of pulsating instabilities theoretically predicted to occur in transient spherical diffusion flames using fuel/diluent mixtures that are above a critical Lewis number, at near-limit conditions.

As suggested by the experiment's name, s-Flame features a spherically symmetric diffusion flame, which is generated by discharging a fuel gas from a porous spherical burner into a still oxygen/inert atmosphere in microgravity. The one-dimensional nature of the system facilitates data acquisition, data reduction, and computational simulation.

The space flight program consists of two main parts: (1) Mapping and characterization of the motion (e.g., expansion) of spherical diffusion flames subsequent to ignition and at extinction because of the finite-rate chemistry and radiative heat loss. The study is focused on the thermal radiation, the adequacy of the detailed chemistry and transport submodels used in computational modeling, and the development of flame front instabilities that could modify the extinction boundaries. (2) Studies of the response of characteristically sooty flames to ignition, the onset of soot in the expanding flames, the soot radiation, and the soot's effects on the flame spread, extinction, and instability.

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the natural flicker of flames, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Characteristic length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion systems, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. For example, chemi-ion species, electronically-excited chemiluminescent species, are at very low concentration in typical flames, making the reduced buoyancy environment valuable for evaluating the minor species chemistry. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

II. Methods/Research Operations

A. Discussion of Method/Protocol

The s-Flame investigation and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the tests were all conducted in a nominally automated mode using pre-programmed scripts. As expected, no s-Flame tests were conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. Thus far, that has been the case for all ACME tests, with the mundane exception of 'manually' halting the test after the flame has extinguished or perhaps if a temperature has reached a critical value.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

A total of 276 flames were ignited for the s-Flame experiment, with 29, 116, and 131 respectively ignited during Increments 60, 61, and 63. These numbers neglect some flames which quenched promptly after ignition. In this regard, science testing began at the end of Increment 60 and the first round of tests were completed during Increment 61. No tests were conducted in Increment 62 due to a lack of crew time, where the second round of tests were fully conducted during Increment 63.

The tests were conducted in two rounds, where the first was conducted in increments 60 and 61 featured 145 flames (nominally) in which the fuel and oxygen (i.e., in the chamber atmosphere) were diluted with nitrogen. The second round conducted in increment 63 featured 131 flames (nominally) in which the fuel and oxygen were instead diluted with helium.

The s-Flame experiment does not have living subjects like a life science experiment, but the tests are independently conducted with various gaseous fuels, including hydrogen,

methane, hydrogen/methane blends, and ethylene. These are simple hydrocarbon gases for which the chemistry is relatively well understood.

No samples (e.g., of soot) were collected for s-Flame; all of the ISS results were electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the flames ignited for s-Flame, including a limited overview of the test conditions, is found in Table 1.

Table 1. Flames ignited for the s-Flame experiment

Test Day[1]	Year	Month	Date	Atmosphere[2]	Fuel bottle	Flames ignited[3]
First round of tests, with nitrogen-diluted fuel and oxygen						
19254	2019	9	11	~21% O_2 with N_2	50% C_2H_4 with N_2	13
19256			13			16
19304		10	31		45% CH_4 with N_2	9
19308		11	4	~30% O_2 with N_2		21
19352		12	18	~21% O_2, with N_2	100% H_2	10
20006	2020	1	6		44.4% H_2 with CH_4	17
20013			13		66.7% H_2 with CH_4	22
20024			24		100% H_2	16
20029			29		50% C_2H_4 with N_2	21
First round of tests, with helium-diluted fuel and oxygen						
20132	2020	5	11	~21% O2 with He	25% H2, 20% CH4, with He	7
20136			15		20% H2, 25% CH4, with He	11
20142			21	~30% O2 with He	30% H2, 15% CH4, with He	17
20155		6	3	~30% and ~21% O2 with He	45% H2 with He	9
20160			8		45% CH4 with He	12
20164			12	~30% O2 with He	20% C2H4 with He	11
20170			18	~30% and ~21% O2 with He	25% C2H4 with He	20
20176			24		30% C2H4 with He	23
20190		7	8		45% H2 with He	21

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Atmosphere** is ~1 atm but pressure increases and O2% drops with vitiation.
[3]**Flames ignited** is the number of nominally sustained flames ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

While there were certainly issues with testing discussed in the next section, there were no significant anomalies during s-Flame's Increment 63 testing. However, anomalies did occur during Increment 60 and 61 which led to corrective actions and/or make-up testing.

2. Any other difficulties in obtaining scientific data

Ignition of the flames presented a significant challenge to the experiment during this increment. All of the flame tests were conducted where the fuel and chamber atmosphere were both diluted with helium rather than the nitrogen used in Increments 60 and 61. The two gases' properties are dramatically different making comparison testing very desirable for manifestation of different phenomena. Helium's thermal conductivity is notably several times greater than that of nitrogen, so considerably more energy and power was required to ignite the flames. The power level was slowly increased from test to test until ignition was finally achieved on the ninth test, where only seven flames were ignited of the 15 tests conducted on May 11 (20132), the first day for flames in Increment 63. The cautious approach was taken because of concerns that the igniter tip might fail (i.e., burn out) if too much power was applied. An igniter failure would have likely halted testing for weeks given the need to schedule the commander's time to replace the igniter tip.

After that first test day, the decision was made to have the igniter replaced with a coil rather than loop design which was believed to be more robust because of the thicker wire. Furthermore, the igniter tip had not been replaced in months, and the tips do degrade with time, especially when they are not quickly retracted from the ignited flames. The coil igniter was used successfully but still with caution, where flames were not always ignited successfully. Subsequently, some tests were conducted in atmosphere with ~30% oxygen (by volume) instead of the planned ~21%. As one would expect, that proved to be helpful for achieving ignition. But the end result was that some tests were conducted with 21% oxygen, some at 30%, and some at both nominal concentrations. Fortuitously, this mixture of conditions may provide deeper insight into the mechanisms of flame dynamics and response.

Tests are also unsuccessful when the flame fails to ignite, where that occurred, for example, during Increment 61 in tests 19304A4, 19304A6, 19308C1, and 20006A1. That can be corrected by adjusting various test parameters and other settings. In this case, adjustments were subsequently made to the ignition position (e.g., after 19304A4 and 19304A6), the relative timing of the fuel flow and igniter activation (e.g., after 19308C1), and the duration of the igniter activation (e.g., after 20006A1). As another example, failed ignition can result from false ignition detection and deactivation resulting from the combination of settings for ignition and detection, e.g., the gain for the Photomultiplier Tube (PMT) module measuring the OH* chemiluminescence. The latter, in the ultraviolet at 310 nm, is a good indicator of an

ignited flame rather than the glowing igniter, which emits primarily in the infrared and visible spectrum.

Testing on June 3 (20155) was cut short due to a combination of issues. The final test was aborted due to an error with an Image Processing and Storage Unit (IPSU). Consoles needed to be reset for the new increment, and there was an issue with the Fuel Oxidizer Management Assembly (FOMA) Control Unit, i.e. the FCU. Rebooting the FCU inadvertently set the valve safety timers to zero, which cannot be reset without venting and refilling the combustion chamber with the intended atmosphere. Those issues combined with an extended loss of commanding prevented further science operations on that day. Therefore, an additional test day (July 8, i.e., 20190) was added to make up for lost science, e.g., where only one test was successfully conducted at ~21% oxygen on June 3. The FCU was off nominal because of a high carrier gas pressure in the Gas Chromatograph Instrumentation Package (GCIP). Imaging of the hydrogen flames was often an issue, where they were not apparent or hardly visible to the operations camera and ACME DataCam, where the latter is the primary camera for ACME testing. The testing with the extremely dim flames was limited to that conducted on Dec. 18 (19352), Jan. 24 (20024), June 3 (20155), and July 8 (20190). This shortcoming unfortunately inhibits assessment of the three-dimensional flame symmetry as the two intensified cameras were opposite of each other.

The flow ranges of the selected Mass Flow Controllers (MFCs) was found to be a challenge, where it was discovered that the goal of studying both kinetic and radiative extinction was not always satisfactorily achieved because of flow rate constraints.

At high flow rates, the burner does not produce spherically symmetric flames. While nominally spherical at moderate flow rates, the flame's center becomes increasing offset from the burner center. More specifically, the flame is shifted toward the supply tube indicating that the sintered attachment of the porous sphere to the supply tube is more 'open' than the pores of the spherical burner. In other words, the flow from the burner is not spherically symmetric. At sufficiently high flow rates, the flame becomes obviously non-spherical, where it can appear to be ellipsoidal or even parabolic in shape. In the latter case, the flame locally quenches near the supply tube.

When at a small fraction of the maximum flow rate, Mass Flow Controller (MFC) performance has been seen to be poor during s-Flame testing, i.e., where actual flow rate differed from that desired. However, this was not a surprise to those familiar with the hardware.

Flow perturbations have been noticed in some s-Flame tests, where ACME's MFCs used to control the flow of fuel, nitrogen, and oxidizer have unfortunately not always performed satisfactorily. In tests conducted using the 1-slpm fuel MFC (where the flow range is on a nitrogen basis), the flow has been seen to overshoot and undershoot when first opening. Small flow irregularities have been most apparent

through their effect on the Photomultiplier Tube (PMT) measurements, e.g., of OH* and CH* chemiluminescence. A momentary increase in fuel flow, as has occurred in some s-Flame tests, naturally leads to increased light from the flame. The MFCs are tested - without flames - after they are first installed. Adjustments are typically made to (1) the setting that controls when the valves open and (2) the Proportional-Integral-Derivative (PID) settings. This ACME activity has involved significant testing, where there is an assortment of MFCs with different flow ranges for each of the three gas types. While the MFC adjustments have been beneficial, they have not eliminated all gas flow problems. It was seen in prior ACME testing that nitrogen oscillations (if any) tend to damp with time, so a nitrogen pre-flow can be used so that such oscillations might end before the flame is ignited. As another measure, the MFCs are re-zeroed prior to each test, where that was another measure adopted in earlier ACME testing.

As mentioned previously, s-Flame tests have sometimes automatically aborted (because of the controlling software script), where that is normally prior to gas flow and the activation of the igniter – which is typical for other ACME experiments. As a result, there was little lost other than the available experimental time in the failed test attempt. ACME aborts normally occur because of communication issues with an IPSU, which for example happened for tests 19304A5, 19352D3, 19352P1, 20024N1, and 20024J1. A few other tests aborted because flow rates were accidentally specified outside of the range of the installed mass flow controller(s), where that included for example 19352A1, 19352D2, and 20013C2. ACME aborts are infrequent, but there are sometimes days when they occur repeatedly as can be seen here.

C. Completeness/Quality of Data

During Increments 60 and 61, nine days of tests were conducted during which 145 flames were ignited for the first round of testing. In contrast, only 29 tests were specified as required or highly desired in the ACME Integrated Science Requirements Document (ISRD) for that round. Therefore, a wealth of data was obtained for this round of testing which featured gaseous fuels diluted by nitrogen.

However, there was a significant loss of data. Most importantly, the first fuel bottle appeared to leak on installation (in Increment 60) and later on-orbit testing (in Increment 61) with other hydrogen/methane blends suggests that its fuel mixture would have been ideal for achieving science success. Furthermore, there was significant image data loss on two different test days, specifically Sept. 13 (19256) and Dec. 18 (19352). As a result, those test days were nominally repeated on Jan. 29 (20029) and Jan. 24 (20024), respectively (i.e., in a reversed order). For these reasons, a simple comparison of the total number of flames ignited (145) with that specified as required or highly desired (29) for the first round of testing is somewhat misleading.

As discussed in the preceding section, the hydrogen and diluted-hydrogen flames were not apparent or extremely dim in imaging from the operations camera and ACME DataCam. Neglecting that factor, the cameras are sensitive to radiation where pixels

gradually become 'hot' (i.e., dead) over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this; and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS).

D. Status of Data Analysis **[provided by Prof. Stephen Tse, Rutgers U.]**

Image files are currently being processed. Radial positions of maximum CH* chemiluminescence is being plotted against time. Different radii are being considered and plotted together, as some flames are not completely symmetric. MFC and PMT data are being compared with video images to assess and confirm experimental settings. Radiometer data is being reviewed. Analyses of the fiber pyrometry data are ongoing. MATLAB code is used to isolate fiber images, which are then color assessed for temperature.

Computational simulations with detailed chemistry and transport of the test cases are underway. Time-dependent simulations will be compared with the time-dependent experimental data, where tracking of the flamefront location is based on chemiluminescent emission. Steady-state simulations will produce the Isola curves mapping flame response versus system Damkhöler number for given fuel mixtures. The experimental cases will then be pin-pointed on the computational mappings to assess whether the experimental conditions are in quasi-steady state or near extinction limits, either kinetic or radiative. Oscillation behavior will be correlated to possible pulsating oscillations found in the simulations.

E. Preliminary Research Findings **[provided by Prof. Tse and lightly edited by Stocker]**

The s-Flame experiment utilizes spherically-symmetric diffusion flames and their dynamics and responses to investigate fundamental processes, such as extinction, instability, and soot formation.

The tests conducted on 19254 (near the end of Increment 60) revealed that a quasi-steady flame could be attained for extended duration at low flow rate, i.e., 3 cc/s for 20% C_2H_4/80% N_2 (on a volume basis). The flame was shut down by the automated system because the burner reached a cut-off temperature of 250°C. Although the flame oscillated near the end of the test, it was likely due to influence of the burner tube. Comparisons with computation simulations will help to clarify the involved processes. For higher flow rates, e.g., 5-10 cc/s, with the same mixture, radiative extinctions were observed. Generally, the flames showed unstable behavior near the burner tube prior to extinction, likely because of lack of fuel mixture and heat loss in that region. Other mixtures, e.g., 25% C_2H_4/75% N_2 and 30% C_2H_4/70% N_2, also showed radiative extinctions at higher flow rates. For some cases, the pyrometry fiber interacted with the flame, allowing determination of temperature in on-going analyses. Soot was observed only

during ignition, which went away as the flame established itself. To assess whether these nitrogen-diluted ethylene flames are non-sooty or sooty, cases were run where mixtures were ignited and established at low flow rates (e.g., 3 cc/s) with low ethylene contents (e.g., 25-30% C_2H_4) and then ramped to high flow rates (e.g., 5-10 cc/s) with higher ethylene contents (e.g., 40-50% C_2H_4). The flames all radiatively extinguished, but no sooting was observed. Observations of such radiative extinction behavior at high flow rates is an achievement because such phenomenon was not observed in our ground-based studies, given the limited microgravity durations. Radiative reabsorption extends flame size at extinction. In ISS testing, radiative extinction has been found to be a common event, which was predicted computationally, but not observed previously for our flame conditions.

The 19256 tests (in Increment 60) were aimed at examining kinetic extinction. Similar to the radiative extinction cases, sufficient microgravity duration is needed to observe kinetic extinction because a near steady flame needs to be established first, before ramp down in flow rate to find properly the chemical kinetic extinction limit. High flow rate cases (e.g., 13.5 cc/s, 15 cc/s) showed asymmetrical flames which radiatively extinguished. Flow rates were lowered so that quasi-steady flames could be established before ramp down in flow rate. The 15% C_2H_4 mixtures were not strong enough to sustain a quasi-steady flame at 3 cc/s before ramp down. However, for stronger mixtures, i.e., 20-30% C_2H_4, very nice flames were established at 3 cc/s; and kinetic extinctions were observed during ramp down before reaching 1 cc/s. Radiative extinctions for ramp up cases were also examined. Similarly, for stronger mixtures, i.e., 20-30% C_2H_4, very nice flames were established at 3 cc/s; and radiative extinctions were observed during ramp up before reaching 8 cc/s.

Testing on 19304 investigated 45% CH_4/55% N_2 mixtures issuing into an air atmosphere. The results indicated that the flow rates specified by the fuel MFC (0.5 slpm) were in the regime of kinetic extinction. Quasi-steady flames could not be established. Furthermore, larger flow rates could not be provided by the MFC, so observations of radiative extinction could not be achieved. Even with a larger capacity MFC, characteristically larger flames would be expected to be asymmetric. As such, nitrogen-diluted methane flames were planned to be reexamined in increased oxygen atmosphere, i.e., 30% O_2/70% N_2.

Consequently, the 19308 tests investigated nitrogen-diluted methane flames in 30% O_2/70% N_2 atmosphere. Various methane concentrations from 45% down to 15% were examined. For 45% CH_4/55% N_2, relatively long duration flames of ~15 s and ~10 s were established at flow rates of 2 cc/s and 3 cc/s, respectively. At larger flow rates of 5 and 9.4 cc/s, the flames expanded to large radii and then extinguished. These flames appeared to demonstrate radiative extinction. Reduction of flow rate from 2 cc/s to 1 cc/s resulted in extinction during ramp down, indicating kinetic extinction. For 30% CH_4/70% N_2, flow rates of 3, 5, and 10 cc/s produced flames that expanded to extinction (assumed radiatively); while reduction of flow rate from 3 to 1 cc/s resulted in presumably kinetic extinction during ramp down. For 25% CH_4/75% N_2, flow rates of 5 and 10 cc/s

yielded flames that expanded to extinction. For 20% CH_4/80% N_2, constant flow rate of 10 cc/s created a flame that expanded until extinction (assumed radiatively); while reduction of flow rate from 10 cc/s to 1 cc/s resulted in wavering extinction during ramp down, suggesting kinetic extinction. For 15% CH_4/85% N_2, flames were ignited, but extinguished very quickly. These tests of 19308 suggest that they could manifest both radiative and kinetic extinction for the conditions specified.

The 19352 tests examined various concentrations of H_2 (45%-70%) diluted with N_2 issuing into air. The installed fuel MFC had a maximum output of 0.1 slpm, which unfortunately provided a less-than-sufficient flow range where its use may have precluded the study of radiative extinction corresponding to larger flame sizes. Imaging the diluted hydrogen flames was a challenge, especially since the hobj image files from the ACME DataCam were mis-formatted. Furthermore, little could be seen in the analog ops imaging, e.g., where it, like the DataCam and Cool Flames Investigation (CFI) intensified camera, cannot image the ultraviolet OH* flame emission (at 310 nm). Analysis of images from the Low Light Level Ultraviolet (LLL-UV) camera and its OH* filter is ongoing. And as one might expect, little could be seen with the CFI intensified camera using its broadband CH_2O filter. Quasi-steady flames seem to be present at low flow rates (e.g., 2.4 cc/s for 70% H_2), as inferred from PMT, pyrometry fiber emission, and burner temperature observations. Kinetic extinction seems to occur as the flow rate is ramped down. There were various problems with the IPSU and some tests aborted. Using the 100% H_2 bottle with a higher flow rate MFC was later performed for tests of 20024 (see below), which was a make-up test day added because of the mis-formatting of the DataCam image files.

The tests conducted on 20006, using the 44.4% H_2/55.6% CH_4 bottle, examined 20% H_2/25% CH_4/55% N_2 mixtures issuing into air. Flames extinguished, some with wavy oscillating behavior prior to extinction. However, it seems that even for very small flow rates, e.g., 2 cc/s, extinction was by the radiative mechanism during flame expansion. There was too little parameter room in terms of flow rate and burner size to identify conclusively kinetic extinction during ramp down for these mixtures. In short, radiative extinction (with local flame extinction and re-ignition seemed to occur prior to ultimate extinguishment) during flame expansion after ignition was observed. However, it is unclear whether ramping down the flow rate resulted in kinetic extinction. Detailed analyses of the data are needed. The various cameras seemed to capture the tests well.

The 20013 tests, which used the 66.7% H_2/33.3% CH_4 bottle, mainly investigated 30% H_2/15% CH_4/55% N_2 mixtures issuing into air. Radiative extinction was difficult to identify conclusively because flames with larger flow rates, e.g. 15 cc/s, seemed to last, although with patchy extinction areas, until time out. Larger flow rates could not be examined either because of limited MFC range or distorted flame shape cause by burner limitations at high flow rates. Also, kinetic extinction during ramp down could not be identified conclusively, as some flames seemed to extinguish because they were "going into the burner."

The 20024 tests using the 100% H_2 bottle repeated and augmented conditions for those of 19352 where the 0.1-slpm fuel MFC range was too low to examine radiative extinction and the analog and intensified cameras did not pick up the flame images at all or well. Here, the 0.5-slpm fuel MFC was used and only a fused silica filter was used for the Cool Flames Investigation (CFI) intensified camera. 45%-15% H_2/N_2 balance mixtures were investigated. The flames at high flow rates, e.g., 10 cc/s, did not extinguish, appearing quasi-steady until time out. The pyrometry fibers glowed, perhaps making good flame-structure measurements, which will be evaluated subsequently. The flames seemed to extinguish during ramp down, but further analysis is needed to assess kinetic extinction.

The tests conducted on 20029 used 50% C_2H_4/50% N_2 and repeated and augmented conditions of 19256, when the images were not good. Again, the results indicated both kinetic and radiative extinction phenomena under corresponding conditions. However, no sooting flames were established, necessitating 100% C_2H_4 and an enriched oxygen atmosphere to facilitate sooting behavior in possible future tests.

The 20132 tests kicked off a round of tests with helium-diluted fuel and oxidizer. The ambient was 21% O_2 and 79% He. The fuel was composed of 25% H_2, 20% CH_4, and 55% He. Using He as inert made igniting the mixture much more difficult compared to that for the N_2 inert cases of Round 1, using the same ignition parameters. The reason is believed to be due to the 6X higher value in thermal conductivity of He versus that of N_2. As a result, ignition parameters were varied, e.g., with a longer preheat time and higher power, to achieve ignition for the He-diluted mixtures and ambient. The tests produced flames at 3, 5, 7, and 10 cc/s that expanded outwardly quickly and extinguished. For 5, 7, and 10 cc/s, the flames seemed to disappear and 're-ignite' with a "flash" before final extinction. We are currently investigating the flash phenomenon. Moreover, the radiation mechanism is not conclusive, as it appears different than for the N_2-diluted cases. We did not have a chance to examine lowering of flow rates to assess kinetic extinction.

The 20136 tests examined 20% H_2/25% CH_4/55% He issuing into 21% O_2/He. For 5 cc/s and 7 cc/s, the mixture ignites but immediately extinguishes, with almost no observable flame-front expansion. For 10, 13, 15, and 17.5 cc/s, the mixtures ignite, expand quickly, and extinguish. We believe that the extinction mechanism is radiative and will compare with computational simulations. For the cases with a ramp down in flow rate, the flames seemed to extinguish before ramp down. Thus, we were not able to assess purely kinetic extinction. Ignition parameters were varied to investigate thresholds for igniting the mixture for different flow rates. Only flow rates above 7 cc/s showed flame-front expansion leading to extinction, with 17.5 cc/s showing a flash or wavering of the flame before final extinction.

The tests conducted on 20142 investigated 30% H_2/15% CH_4/55% He issuing into 30% O_2/He. Elevated oxygen was approved for study, since ignition and extinction limits in previous testing were affected by the high thermal conductivity of He. For 5, 7, 10, 15,

13, and 16.5 cc/s flow rates, nice flames were established that expanded to wavy/oscillating extinction at the end, indicative of radiative extinction. For the cases with a ramp down in flow rate, the flames seemed to extinguish during the ignition flow rate stage before ramp down. Thus, we were not able to assess kinetic extinction.

The 20155 tests examined 45% H_2/55% He issuing into 30% O_2/He and 21% O_2/He. Previous tests using He as inert had shown that much larger energy and power are needed to ignite the mixtures. For the 30% O_2/H_2 atmosphere, the tests produced flames at 10 and 14 cc/s (MFC limit of 1 SLMP) that expanded after ignition, but were extinguished by the automated protocol because the burner reached the 250°C temperature limit before any self-extinction could occur. The flames were not too large at the end. Presumably, the thermally-conductive He diluent facilitates heat transfer to the burner. For the cases with a ramp down in flow rate, i.e., starting at 10 and 14 cc/s and ramping down, the flames seemed to self-extinguish during ramp down. Flow rate at 5 cc/s results in flames that only existed briefly. Extinction at this lower flow rate is probably due to heat loss to the burner, rather than pure kinetic extinction. Ramp down from 14 cc/s towards 5 cc/s resulted in the burner reaching the 250°C limit during ramp down. More analyses are needed. Tests at 21% O_2/He atmosphere were limited. Fuel flow of 5 cc/s showed flash ignition, but no sustained flame. Meanwhile, 10 cc/s resulted in a flame that seemed to expand to self-extinction.

Testing on 20160 investigated 45% CH_4/55% He issuing into 30% O_2/He and 21% O_2/He. Without H_2 in the mixture, even larger energy and power were needed to ignite the mixtures. For the 30% O_2/H_2 atmosphere, the 5, 10, and 15 cc/s flow rates were sooty only at ignition; and the subsequent non-sooty flames expanded to self-extinction. At 3 and 4 cc/s, the mixtures flashed at ignition, and extinguished almost immediately. It was not clear whether extinction was due to kinetic extinction or heat loss to the burner. At 6 and 7 cc/s, flames ignited with less soot, expanding with no soot to self-extinction. The case with a ramp down in flow rate could not conclusively identify kinetic extinction. More analysis is needed. Tests at 21% O_2/He atmosphere needed even more energy and power for ignition. At 5, 7, 10, and 15 cc/s, flames ignited with soot, and expanded quickly to extinction.

The 20164 tests began study of ethylene-based flames, where 20% C_2H_4/80% He issued into 30% O_2/He and 21% O_2/He atmospheres. For 5 cc/s, the flame expanded and remained quasi-steady until the burner limit temperature (250°C) was reached. At 6, 7, 10, and 15 cc/s, the flames expanded to extinction, with the higher flow rates (i.e., 10 and 15 cc/s) exhibiting sooty ignition. Fuel flow of 5.5 cc/s seemed to correspond to a condition where the onset of oscillating extinction was simultaneous with the burner reaching the 250°C temperature limit. For the cases with a ramp down in flow rate, the flames seemed to extinguish during ramp down, but heat loss to the burner may be the mechanism of extinction. Although 13 tests were conducted with the 21% O2/He atmosphere, ignition was not achieved.

The tests performed on20170 investigated 25% C_2H_4/75% He issuing into 30% O_2/He and 21% O_2/He. At 5 and 5.5 cc/s, nice blue chemiluminescent flames were ignited, but eventually were cut off because of the burner temperature limit being reached. At 6 cc/s, a nice blue flame was ignited which expanded slowly to extinction. At 7, 10, and 15 cc/s, sooty ignition occurred, but as the flames expanded, the soot went away, eventually resulting in radiative extinction. By ramping the flow rates up, starting at 5 cc/s, soot at ignition was minimized, and the flames could expand to extinction. Ramp down of the flow rate resulted in almost immediate extinction. For tests at 21% O_2/He atmosphere, cases at 5, 10, and 15 cc/s resulted in flames that expanded quickly to extinction. Less soot was present at ignition compared with that for the 30% O_2/He atmosphere cases. Consistent ignition was not always achieved for the same conditions, which is puzzling.

Testing on 20176 explored 30% C_2H_4/70% He issuing into 30% O_2/He and 21% O_2/He. At 5 cc/s, a long duration flame was established that eventually reached the burner temperature limit. At 6, 7, 10, and 15 cc/s, the flames were sooty at ignition, but the soot dissipated; and they expanded to extinction. Ramping up the flow rates, starting at 5 cc/s, minimized soot at ignition and resulted in flames that slowly expanded to extinction. For tests at 21% O_2/He atmosphere, cases at 5, 10, and 15 cc/s resulted in flames that expanded to extinction. At 4 cc/s, the flame lasted a while before eventual extinction. The 3 cc/s case did not ignite. Ramp down of the flow rate seemed to result in flames that extinguished quickly before the protocol could ramp down the flow rate.

The last testing of Round 2, which was a makeup of 20155, occurred on 20190, with 45% H_2/55% He issuing into 30% O_2/He and 21% O_2/He. For both oxygen concentration environments, flow rates resulting in expanding flames lasted for a while, but were cut off by automated protocol before self-extinction when the burner temperature limit was reached. As for cases where the flow rates were ramped down, extinction detection was observed for both oxygen concentration environments. We are currently analyzing the data to determine if extinction was kinetic or due to heat loss to the burner. A last set of tests were conducted with a reduced oxygen concentration atmosphere, i.e., 17.5% O_2/82.5% He. Some cases indicated that radiative and kinetic extinction were possibly observed.

In summary, selected preliminary findings from this testing include:

- Cases run for nitrogen-diluted methane and ethylene flames and in air seemed to indicate kinetic extinction at low flow rates and radiative extinction at high flow rates. As previously mentioned above (and in our report for the preceding increment pair), such phenomena were not observed in our ground-based research, given the limited microgravity durations. Nevertheless, comparisons with computational simulations are needed to assess better the extinction behavior.
- Cases run for nitrogen-diluted hydrogen flames seemed to show kinetic extinction at ramp down. However, for the mass flow rates that could be accommodated, radiative extinction was not observed.

- Cases run for nitrogen-diluted hydrogen/methane mixture flames manifested extinction, which are presumed to be radiative. However, kinetic extinction at low flow rates could not be identified. More analyses of the data are needed.
- Cases for helium-diluted cases shifted the extinction limits with respect to the nitrogen-diluted cases. In general, kinetic extinction was difficult to observe for the conditions examined. More analyses are needed to assess instability near the flame limits, which was predicted to occur computationally.

Early ISS results were presented in the following presentations:

1. Chung K. Law and Stephen Tse, "ACME - s-Flame," ISS Increments 61 & 62 Science Symposium, 17-19 Sept. 2019. (oral presentation)
2. Stephen D. Tse, Jonathan Shi, Delin Zhu, and Chung K. Law, "Dynamics of Burner-Generated Spherical Diffusion Flames In Microgravity," 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, 20-23 Nov. 2019. (poster presentation)
3. Jonathan Shi, Stephen D. Tse, Dennis Stocker, and Chung K. Law, "ISS Experiments of Gaseous Spherical Diffusion Flames," 36th Annual Meeting of the American Society for Gravitational and Space Research, 5-6 Nov. 2020. (pre-recorded oral presentation)

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Video(s) describing the ACME project can be found at https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/.

Videos related to s-Flame and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

Product references are included for clarification and do not indicate an endorsement by NASA or the federal government.